KOFORIDUA TECHNICAL UNIVERSITY

FACULTY OF ENGINEERING

ELECTRICAL/ELECTRONIC ENGINEERING DEPARTMENT

MAIN TOPIC

DESIGN, FABRICATION AND TESTING OF A SEMI-AUTOMATED GLASS CRUSHER MACHINE FOR RECYCLING GLASS FOR INDUSTRIAL USE

PROJECT TOPIC

MOTOR CONTROL AND SOFTWARE IMPLEMENTATION

BY:

STUDENT NAME:	INDEX NUMBER:
Edwin Setsoafia Kwadzo	B302210013
Joseph Nkansah	B302210002
Titus Cabutey Kartey	B302210029
Kelvin Nongor Borketey	B302210011
Christian Akuffo	B302210012
Manasseh Aboagye Yeboah	B302210024

A PROJECT WORK SUBMITTED TO THE DEPARTMENT OF ELECTRICAL/ELECTRONIC ENGINEERING, FACULTY OF ENGINEERING IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF BACHELOR OF TECHNOLOGY CERTIFICATE IN MECHATRONICS ENGINEERING

May, 2025

STUDENT'S DECLARATION

We hereby declare that this project work submitted is a record of an original work done by the group under the supervision of Mr. Richard Mensah of the Electrical/Electronic Department, Faculty of Engineering. The results embodied in this project have not been submitted or presented for another certificate in this institution or elsewhere.

Edwin Setsoafia Kwadzo	....	....
(B302210013)	DATE	SIGNATURE
Joseph Nkansah	....	....
(B302210002)	DATE	SIGNATURE
Titus Cabutey Kartey	....	....
(B302210029)	DATE	SIGNATURE
Kelvin Nongor Borketey	....	....
(B302210011)	DATE	SIGNATURE
Christian Akuffo	....	....
(B302210012)	DATE	SIGNATURE
Manasseh Aboagye Yeboah	....	....
(B302210024)	DATE	SIGNATURE

SUPERVISOR'S CERTIFICATION

I hereby certify that this project work was supervised in accordance with the University's guidelines for supervision of project work.

………………………………	……………………………
(Supervisor's Name)	(Signature/Date)
………………………………	……………………………
(Head of Department's Name)	(Signature/Date)

DEDICATION

We wish to dedicate this project to the Mechatronics Engineering teaching staffs under the Electrical/Electronic department of the Koforidua Technical University for their invaluable support from the commencement, to the final phase of this academic research project. We would like to thank the entire Electrical/Electronic department for giving us the opportunity to apply the various acquired knowledge and engineering skills in a team environment helping us shape our team-work abilities to the core. We would also like to narrow down our sincere appreciation to our project supervisor Mr. Richard Mensah for his guidance in each phase of this project. Eventually, we dedicate this project to ourselves – the Motor Control and Software Implementation Team and the other remaining three (3) teams that made this project a success for their team-work, countless disagreements (which shaped this project during the initial stage), and not forgetting the sleepless nights of research work.

ACKNOWLEDGEMENT

A project of this nature is the result of direct and indirect collaboration and support of many colleagues, families and our various supervisors. Foremost, we want to thank our Almighty God for His goodness, mercy and peace in our lives. Also, we want to thank our affable supervisor, Mr. Richard Mensah for his time, dedication and efforts in guiding us to complete this project. In addition, we extend our deepest appreciation to our families for their immense support, financially and also emotionally. We would like to thank everyone else who has indirectly contributed to this project. We the Motor Control and Software Implementation team says God richly bless you all.

ABSTRACT

This project presents the design and implementation of a semi-automated glass crusher system for glass recycling and industrial use. The primary objective is to develop a motor control and software architecture that ensures precise, real-time control of the crushing mechanism while integrating critical safety features to improve efficiency and operational safety. The system employs event triggered and time triggered operations for task scheduling and real-time responsiveness. A system operating in real-time must be able to provide the correct results at the required time deadlines. This implies that, what is needed must be ready at the time that it is needed. Key components include;

A PID (Proportional-Integral-Derivative)-controlled Permanent Magnet Synchronous Motor (PMSM with current feedback) to regulate crushing force and prevent overload.

Real-time current sensing to prevent system damage due to over-current.

Real-time system-state monitoring for early fault detection and ensuring efficient power consumption.

User interface (LCD, LEDs, buzzer, buttons) for system status, alerts and start/stop/pulse operations enabling manual control and feedback. The chosen firmware architecture adopts a modular design, ensuring robust coordination between motor control algorithm, filtered sensor data and safety interlocks. Preliminary testing of individual components confirms reliable system performance with proactive protection against overloads. This project demonstrates the effectiveness of embedded systems in industrial automation and proposes scalability improvements for broader waste management applications.

List of Figures

List of Tables

Table of Contents

CHAPTER 1: GENERAL INTRODUCTION

1.1 Overview

"A crusher is a machine that is designed to reduce large solid masses of raw material into smaller portions. Crushing is the process of transferring a force amplified by mechanical advantage through a material made of molecules that bond together more strongly, and resist deformation more than those in the material being crushed do" [1]. In this research work, the material being crushed refers to glass waste such as bottles and broken pieces of glass. A collaboration between the various divisions of the final year Mechatronics Engineering class of 2025 to improve waste management and recycling of the aforementioned glass materials is currently underway. The divisions include the following teams:

Mechanical Systems Design team, responsible for synthesizing mechanisms, designing mechanical components like shafts, gears, linkages and other mechanical parts which shall together form the mechanical sub-assembly of the overall system. The Mathematical Modelling and Control team is responsible for mathematical modelling and design of a suitable mathematical controller for the system. Electronics and Communication team responsible for designing and or selecting electronic sensors, board to board communications systems, Printed Circuit Boards (PCBs), and all other electronics related tasks. As the Motor Control and Software Implementation team, our work is centered on designing and implementing control algorithms and other software related controls for the smooth and safe operation of the entire system. This documentation further details our research work.

1.2 Background

Glass waste is a significant environmental issue, especially in urban and industrial areas where large quantities of bottles, jars, and other glass containers are discarded daily. While glass is 100% recyclable without loss in quality or purity, improper disposal and accumulation in landfills hinder effective recycling. Manual crushing of glass for recycling is labour-intensive, unsafe, and inefficient. The increasing global emphasis on sustainable waste management has highlighted the critical need for efficient glass recycling systems. Traditional glass crushing methods often rely on manual operation or rudimentary mechanical processes, which present significant limitations in terms of safety, energy efficiency, and output consistency [2] [3]. Hence, the need arises for a semi-automated glass crushing system. This project aims to contribute to sustainable waste management by developing a motor-controlled glass crushing machine. The focus is on designing and implementing a motor control system that can automate the crushing process, enhance safety, and improve efficiency. This aligns with modern engineering goals of automation, energy efficiency, and environmental preservation. The motor control of the physical system is very much significant in achieving the project objectives. Two Permanent Magnet Synchronous Motors (PMSM) with gears has been selected for the crushing mechanism. Also, an Induction Motor has been incorporated for the manual conveying of the raw glass waste into the crusher. PMSM is preferred over other motors because of their high torque density with exceptional dynamic performance and efficiency. With their robust features which include high starting torque, quick response under field-oriented/vector control, high energy efficiency and regenerative braking capability makes them useful in applications like robotic manipulators, industrial servo systems, electric and guided vehicles and high-performance automation where precise, responsive speed control is required. In crushing applications, the integration of real-time motor control with comprehensive safety monitoring has proven effective in preventing equipment damage and operator hazards. The development of such systems requires careful consideration of several technical aspects. The crushing process demands precise torque and speed regulation to handle varying glass thicknesses while preventing motor overload. PID control algorithms, when combined with current feedback, have become the industry standard for such applications. Environmental factors such as temperature fluctuations and mechanical vibrations can significantly impact system performance. Implementing proper signal conditioning through low-pass filtering techniques ensures accurate readings from current sensors. Industrial automation systems require multiple layers of protection, including hardware interlocks (emergency stops) and software safeguards (current monitoring, thermal protection). These measures are particularly crucial in user environments where operational safety is paramount. For the User Interface (UI) design, effective human-machine interaction is essential for system monitoring and control, especially in a setting where the equipment may be operated by individuals with varying technical expertise. This project builds upon these technological foundations to develop a semi-automated Glass Crusher that addresses the specific needs of manual non-automated glass crushing systems.

1.3 Problem Statement

The traditional method of glass waste disposal involves collection and storage, often followed by crushing (normally not automated). This method is inefficient for handling large volumes of glass. Current industrial solutions are often expensive and not suitable for small scale applications or local recycling businesses. There is a need for a cost-effective, safe, and efficient solution to automate the glass crushing process, particularly focusing on a reliable motor control system to drive the mechanism.

1.4 Aim of the Project

To design and implement an embedded firmware incorporating motor control and system monitoring algorithms for a semi-automated glass crushing machine.

1.4.1 Objectives

1. To develop and implement a user interface for real-time system monitoring feedback, user control inputs and I2C serial communication protocol.
2. To develop and implement soft start/stop/pulse operations for smooth running of the crushing motor.
3. To perform test procedures in a form of proof-of-concept protype to ensure the firmware runs smoothly before deployment and integration into the main crusher.

1.5 Scope of the Project

This thesis focuses on the design, fabrication and testing of a Semi-automated Glass Crusher tailored for local or small businesses in glass recycling and industrial use.

1.5.1 Scope Limitations

While the project involves various aspects of the Glass Crusher development, the following aspects are considered outside the scope of this thesis:

1. Hardware Design Beyond Motor Control: The design of physical structures, crushing mechanism, sieving mechanism, conveyor motion, circuit enclosures, electronic and electrical circuits, and mechanical components like fasteners and gears will be considered based on existing infrastructure and determined in collaboration with their respective teams on this project.
2. Advanced Safety Features: While basic safety measures will be incorporated, functionalities like sound-dampening housing or vibration isolators for noise reduction and advanced LOTO (Lockout/Tagout) procedures are not the primary focus at this stage.
3. Full Automation: While the entire Glass Crusher System will not be fully automated with an automated crushing process through glass detection, automated discharge and collection of crushed glass particles and automated system logs, there will be an automated feeding system via a conveyor belt, emergency

automation protocols (fail safe mechanisms) and automated system monitoring alerts making the crusher semi-automated.

1.5.2 Deliverables

This project aims to deliver the following key outcomes:

1. A functional Glass Crusher Machine prototype incorporating the developed motor control algorithms and safety measures.
2. A report detailing the design process, implementation details, testing results, and evaluation findings.
3. Recommendations for future research and system enhancements, potentially including a fully automated Glass Crusher with IoT (Internet of Things) capabilities for system data logging.

1.6 National Relevance of the Project

Glass waste forms a significant portion of Ghana's solid waste stream, particularly in urban areas, yet it remains largely unutilized and accumulates in landfills. This project addresses this environmental and industrial gap through a practical, locally engineered solution. The development of a Semi-automated Glass Crushing machine holds substantial national relevance in the following key areas:

1. **Environmental Sustainability:** Ghana faces increasing environmental challenges due to the accumulation of non-biodegradable materials such as glass waste. This project supports sustainable waste management by enabling the conversion of waste glass into reusable cullet. Crushed glass can be repurposed for the production of pavement blocks, tiles, and other construction materials. By promoting eco-friendly recycling practices, the project aligns with the objectives of the National Environmental Sanitation Policy of Ghana.
2. **Support Local Recycling Industry:** The fabrication of an affordable Glass Crusher locally strengthens the capacity of Ghana's recycling sector, particularly among small and medium enterprises (SMEs). It reduces dependency on expensive imported machinery and fosters self-reliance. This project thereby contributes to the growth of a resilient circular economy.
3. **Job Creation and Youth Empowerment:** Widespread adoption of glass recycling technology can create employment opportunities in areas such as waste collection, processing and manufacturing. In addition, this student-led engineering initiative showcases the innovation potential within Ghanaian tertiary

institutions, inspiring youth-led solutions to national challenges and promoting technical entrepreneurship.

4. **Reduction in Import Cost:** Ghana currently imports much of its individual and recycling equipment. This project demonstrates the feasibility of building effective machinery locally, offering a cost-saving alternative that reduces foreign exchange expenditure. It highlights the importance of local development in engineering and manufacturing.
5. **Educational and Research Impact:** This project provides a hands-on platform for we the students in applying mechatronics principles including, mechanical design, control systems, electronics and software integration. It supports interdisciplinary collaboration and contributes to the development of engineering education and applied research within Ghana.

CHAPTER 2 : LITERATURE REVIEW

2.1 Introduction

Glass recycling is a critical component of sustainable waste management, yet traditional manual crushing methods pose significant safety hazards, inefficiencies, and inconsistent output quality. Advancements in automation, embedded systems, and motor control technologies offer solutions to these challenges, aligning with global standards for occupational safety and environmental protection [2]. This literature review explores the key concepts and methodologies relevant to the development of a Motor Control and Software Implementation solution for the semi-automation of the crushing process.

2.2 Motor Control Principles

Motor control is a critical aspect of automated glass crushers, ensuring safe, efficient and precise operation of the crushing mechanism. Geared PMSM with current feedback offers a dynamic control with high torque at low speeds, precise speed control and overload detection and protection. The current sensor provides real-time current data directly proportional to the torque of the crushing motor. A slight change in the magnitude of the measured current causes a change in the magnitude of the torque. The measured current data is used to feed a PID-algorithm which ensures a closed-loop control of the entire crushing mechanism.

PWM (Pulse Width Modulation) principle adjusts motor speed by varying voltage pulses (duty cycle) hence controlling crushing motor's RPM based on glass hardness or unexpected overloads. PID (Proportional-Integral-Derivative), one of mostly used control algorithms for speed control maintains consistent crushing force and prevents motor overload. Proportional(P) responds to current error (speed deviation). Integral (I) corrects accumulated past errors (long-term changes) and Derivative (D) eventually predicts future errors (sudden jams) [4]. By fine-tuning the proportional, integral, and derivative parameters, PID controllers can precisely control motor speed.

2.3 Software Implementation Technique

The firmware architecture for the glass crusher integrates real-time control, signal processing, safety protocols and user interaction using event triggered, time triggered on an embedded microcontroller. [5] emphasizes the significance of adopting such architecture in embedded firmware design for higher performance and efficient power consumption. Time triggered and event triggered programs aid in task scheduling for concurrent operations (motor control, system health check, system states and user interface updates). It also helps the system in prioritizing interrupts for safety-critical functions.

2.4 Integration of Motor Control and Software Implementation

This section details how motor control algorithms and software architecture work together to create a safe, efficient and responsive automated glass crushing system. The integration combines:

1. Hardware Layer: The hardware consists of an I2C bus with one master microcontroller (which coordinates the motor control tasks), 2 slave controllers, a 20x4 LCD (attached to an HMI panel) on the I2C bus and a stepper motor for gate control operations.
2. Framework Layer: Event-triggered and time-triggered architecture using DarkLight APIs.
3. Safety Layer: Real-time monitoring and fail-safe mechanisms.

2.5 Application to Academic Institutions

Automated glass crushers offer educational, research, and sustainability benefits for universities, technical schools and research labs. Teaching and hands-on learning in engineering disciplines such as Mechanical, Electrical/Electronics and Mechatronics. [6] highlighted the application of motor control and software implementation solutions for a VAC (Vehicle Access Control) system designed and fabricated under a learning environment at Koforidua Technical University. This project showcase motor control and software implementation integrated into a different system other than a Glass Crusher. Even though Glass Crushers are not directly applicable in academic institutions as at the time of writing this report, future prospects are that such a system will gain recognition in educational institutions for the purpose of learning as well as waste recycling on campus starting with this research work.

2.6 Gaps in Knowledge and the Current Research Contribution

Most industrial crushers prioritize efficiency over safety, while academic studies focus only on theoretical safety models. This is evident in [2] as it highlights manual crushing risks and lack of automated solutions for small-scale applications. Few studies optimize real-time safety interlocks (vibration and current sensing for low-cost systems). There is also a limited data on dynamic motor control for varying glass types (bottles, sheets, tempered glass). As evident in [7], their research presents a Glass Crusher Machine but does not address PID tuning for mixed-waste scenarios hence fixed-speed motors were used wasting energy on thin glasses. In addition, most research target large factories neglecting modular designs for schools and municipalities or even smaller towns. Fewer current research initiatives, incorporate safety, efficiency and cost into developing state-of-the-art automated Glass Crushers. Therefore, this thesis also aims to bridge this gap by designing, fabricating, and testing a semi-automated Glass Crusher. The project will focus on a cost-effective and efficient system that integrates motor control principles, and embedded software development.

2.7 Conclusion

In conclusion, this literature review reveals that while industrial glass recycling has been extensively studied, critical gaps remain in small-scale, automated crushing systems particularly in safety, energy efficiency and cost-effective implementation. By using motor control algorithms, researchers can create systems that not only enhance safety but also optimize operational efficiency and cost. This work bridges the gap between theoretical research and practical, scalable solutions, offering a blueprint for safer, sustainable glass recycling in non-industrial settings.

CHAPTER 3 : METHODOLOGY

3.1 Introduction

This chapter outlines the design, system architecture and implementation strategies employed to develop the semi-automated glass crusher firmware. Building on the gaps identified in the literature review (Chapter 2) of this research work, the methodology integrates embedded systems engineering, real-time control algorithms, and safety-focused hardware integration to address the challenges of manual glass crushing.

The chapter is structured as follows:

System Design Framework: Overview of the hardware-software co-design approach.
Component Selection: Justification for actuators and microcontroller choices.
Control Algorithms: Implementation of PID motor control, safety interlocks and state machines.
Communication Protocols: Data exchange protocols used between sensors, actuators and microcontroller(s).
Software Architecture: Event-triggered and time-triggered based task scheduling and other frameworks.

3.2 System Design

3.2.1 Hardware Overview

The selection of components is based on their cost and suitability for the automated glass crusher system. The following components were selected:

1. **Actuators**:
   a) **PMSM:** Two (2) Permanent Magnet Synchronous Motors (PMSMs) have been selected for the crushing mechanism. PMSMs with gearbox and current feedback is required for crushing torque and feedback for PID control.
   Motor Specifications:
   Rated Power: 1000W
   Rated Voltage: 48V
   Rated Current: 25±10% with 50±10% instantaneous maximum current.
   Rated Speed: 3000 rpm with 3200 rpm maximum speed.
   Rated Torque: 3.2Nm with 6.4 instantaneous peak torque.
   Line Resistance: 0.08±10% (25℃) Ω

Figure 3.1: ZLTECH 3 Phase PMSM

b) **Stepper Motor and Driver:** For a 2mm thick, 2.3kg diverter, hinge along 395mm edge, and a 2s 190̊ move, the shaft torque required is equivalent 4.85Nm. With recommended safety factors, a stepper motor with holding torque of at least 7.3-9.7Nm has been selected for a reliable direct-drive operation equivalent to 16rpm.

Figure 3.2: Nema34 Stepper Motor

2. **Microcontroller:**

a) **Master Microcontroller:** The STM32F030RTC6 housed on a custom STM32 development board shown in Figure 3.3, serves as the control centre for the entire system. The STM32F030RTC6 microcontroller incorporates the high-performance ARM Cortex™-M0 32- bit RISC core operating at a 48 MHz frequency, high-speed embedded memories (up to 64 Kbytes of Flash memory and up to 8 Kbytes of SRAM), and an extensive range of enhanced peripherals and I/Os. This microcontroller offers standard communication interfaces (up to two I2Cs, up to two SPIs, and up to two USARTs), one 12-bit ADC, up to 6 general-purpose 16-bit timers and an advanced-control PWM timer, which offers a better performance for our PID control algorithm. The STM32F030RTC6 microcontroller operates in the -40 to +85 °C temperature range, from a 2.4 to 3.6 V power supply. A comprehensive set of power-saving modes allows the design of low-power applications. These features make the STM32F030RTC6 microcontroller suitable for a wide range of applications such as application control for our Glass Crusher and user interfaces.

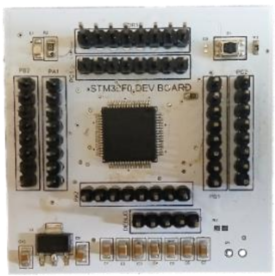

Figure 3.3: DarkLight-V1 Development Board

b) **Slave Microcontroller:** The motor driver boards used in controlling the motor integrates a dedicated microcontroller that functions as a slave unit within the control system which handles motor control tasks. This microcontroller acts as slave; it follows commands from the central master controller. It doesn't make decisions on its own but executes instructions as part of a larger control system.

Pic of motor driver board here

3. **PC2004F - 20x4 LCD (Liquid Crystal Display):** A 20x4 LCD with an I2C PCF8574 interface adapter module was selected as the display for the control unit. From [11] it is obvious that, among the various types of displays outlined, PC1602F and the likes (PC2004F) is the most suitable for our application as no fancy graphics is needed to update the state of the system with sensor readings and user prompts. Below (Figure 3.4) is an image of the 20x4 LCD interfaced with an I^2C module.

Figure 3.4: PC2004F interfaced with an I2C module.

[https://share.google/images/Q1tM4sHgLQ0WQDleI]

4. **Others Hardware:** Figure 3.5, 3.6, 3.7 and 3.8 shows, Momentary non-latching push buttons for start/stop, emergency and pulse operations. Indicator LEDs (Light Emitting Diodes) for system status indication. Buzzer for generating alert sound and an I2C logic level shifter. The logic level shifter connects two different pull-up voltage levels on the I2C bus. 3.3 volts coming from the master microcontroller and 5 volts from the PC2004F LCD.

Figure 3.5: Momentary push buttons

[https://www.lazada.com.ph/products/22mm-waterproof-illuminated-push-button-switch-momentary-switch-with-integral-led-nb5-xb5-aw33b1c-22mm-spring-return-i1816723965.html]

Figure 3.6: LEDs

[https://www.amazon.in/ELECTROPRIME-Yellow-Assorted-Emitting-Diodes/dp/B07L7G3B4R]

Figure 3.7: Buzzer

[https://www.ubuy.com.gh/product/30ABXCG-dc-3v-12v-24v-buzzer-alarm-100db-active-piezo-electronic-buzzer-beep-tone-alarm-ringer-continous-sound-pack-of-2]

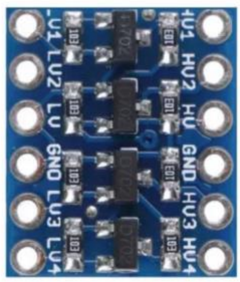

Figure 3.8: 4-Channels, 3.3V -5V Logic Level Shifter

[https://www.robotics.org.za/LEVEL-4P]

3.2.2 Design Architecture and Diagrams

Below we present proposed system architecture and diagrams:

1. Embedded Controller: The entire embedded controller consists of the hardware and software.
2. Hardware Abstraction Layer: The DarkLight APIs uses the Low Level (LL) Drivers to directly communicate with the hardware, abstracting complex bare-metal configurations. Custom Application Programming Interfaces (APIs) such as DigitalOut,

DigitalIn, Wait, Pwm, I2C, that the DarkLight provides abstracts low-level configurations that are time consuming. This APIs are used to generate control signals for the various interfaced peripherals.

3. Event-triggered and Time-triggered: A real-time event-triggered and time-triggered architecture has been chosen to concurrently execute tasks with appropriate assigned priorities by utilizing the APIs as depicted in Table 2. Majority of the tasks like the Display Updater, State Handler, Gate Control, Fault Detector are time triggered tasks that runs concurrently alongside the Motor Control tasks which are external event triggered tasks. All Motor Control tasks have a high priority and therefore are handled by the Interrupt Service Routine (ISR). Communication, Gate Control and Fault Detection tasks have medium priority and are therefore periodic tasks running concurrently in their own thread by the help of the DarkLight Ticker class.

4. Main Program: The main program consists of several functionalities. These functionalities receive input from interfaced sensors and use these inputs to control the actuators of the system as shown in Figure 3.9. A typical illustration would be the current sensors attached to the motor drivers. The current consumption data of the crushing motors is made available on the I^2C bus if requested by the master microcontroller and converted to an equivalent torque value in code after it has been successfully intercepted. This torque value is then fed into the control algorithm, state machine blocks and the display updater task as shown in Figure 3.9. The control algorithm performs PID control on the crushing motors by adjusting the supply voltage to indirectly effect a change in the current torque of the motor using the error value from the previous measured current data, hence maintaining the desired set-point of the crushing motors. The safety interlocks and state machines respectively initiate safety protocols and updates system states on the Human Machine Interface (HMI) panel.

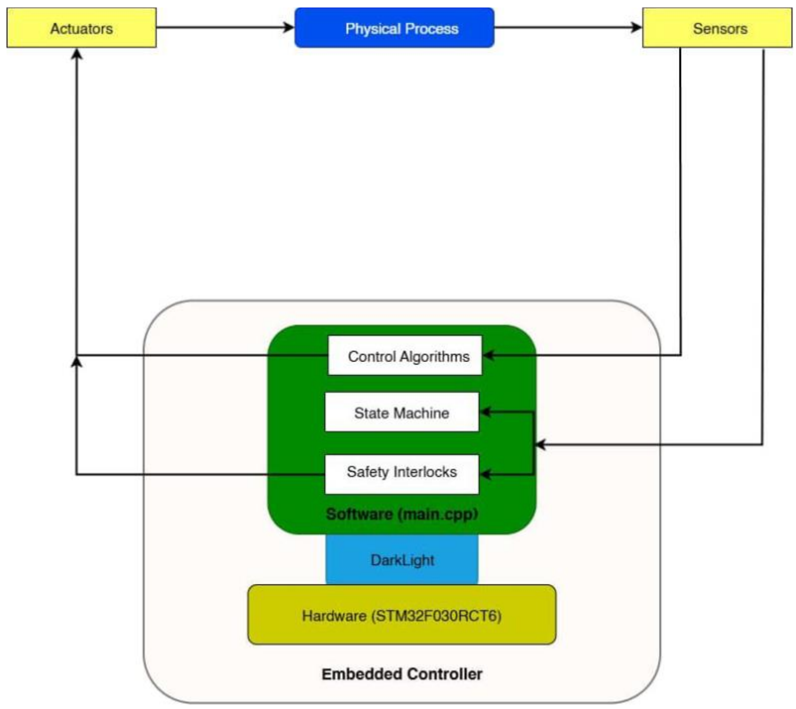

Figure 3.9: System Architecture

TASKS	TASK TYPE	FUNCTION	PRIORITY
Motor Control	Event-Triggered	Real-time interrupt routines for emergency, soft start/stop and pulse operations	High
PID Algorithm	Time-Triggered (Every 50ms)	Control algorithm for overload protection	High
Communication	Time-Triggered	Handles I2C peripheral communications	Medium
Gate Control	Time-Triggered (Every 200ms)	Responsible for opening and closing of gate.	Medium
Fault Detection	Time-Triggered (Every second)	Scans and detects specific fault types on the slave controllers and stepper gate drive	Medium
Display Updater	Time-Triggered (Every 500ms)	Updates LCD displays as system changes state	Low
State Handler	Time-Triggered (Every 200ms)	Scans for the current state of the system and returns it for use in display/motor control/gate control tasks	Low

Table 1: Priority Based Event-triggered/time-triggered system scheduling

3.3 Software Implementation

3.3.1 Programming Environment

The entire firmware for the semi-automated glass crusher is currently being developed in Segger Embedded Studio (SES). Segger Embedded Studio (SES) is a powerful lightweight Integrated Development Environment (IDE) designed specifically for embedded systems development. SES is optimized for ARM Cortex-M & RISC-V, highly efficient compiler based on Clang/LLVM and faster execution and smaller code size compared to GCC-based IDEs. It also comes with a Real-time terminal (RTT Viewer) for live logging without extra hardware. In addition to the RTT Viewer, there is System View a powerful tool for verification and validation. It records, analyses and visualizes captured data in real-time; documents tasks, interrupts, user events. The choice of the programming environment, heavily relied on the above listed advantages of SES. Figure 3.11 shows the project tree of the glass crusher in SES with Segger's System View in use for recording and monitoring user events in real-time. In addition to that, we employed version control tools (Git and GitHub) for remote collaboration, teamwork and version control. This work can be viewed at [https://github.com/ayamEdwin/Glass Crusher].

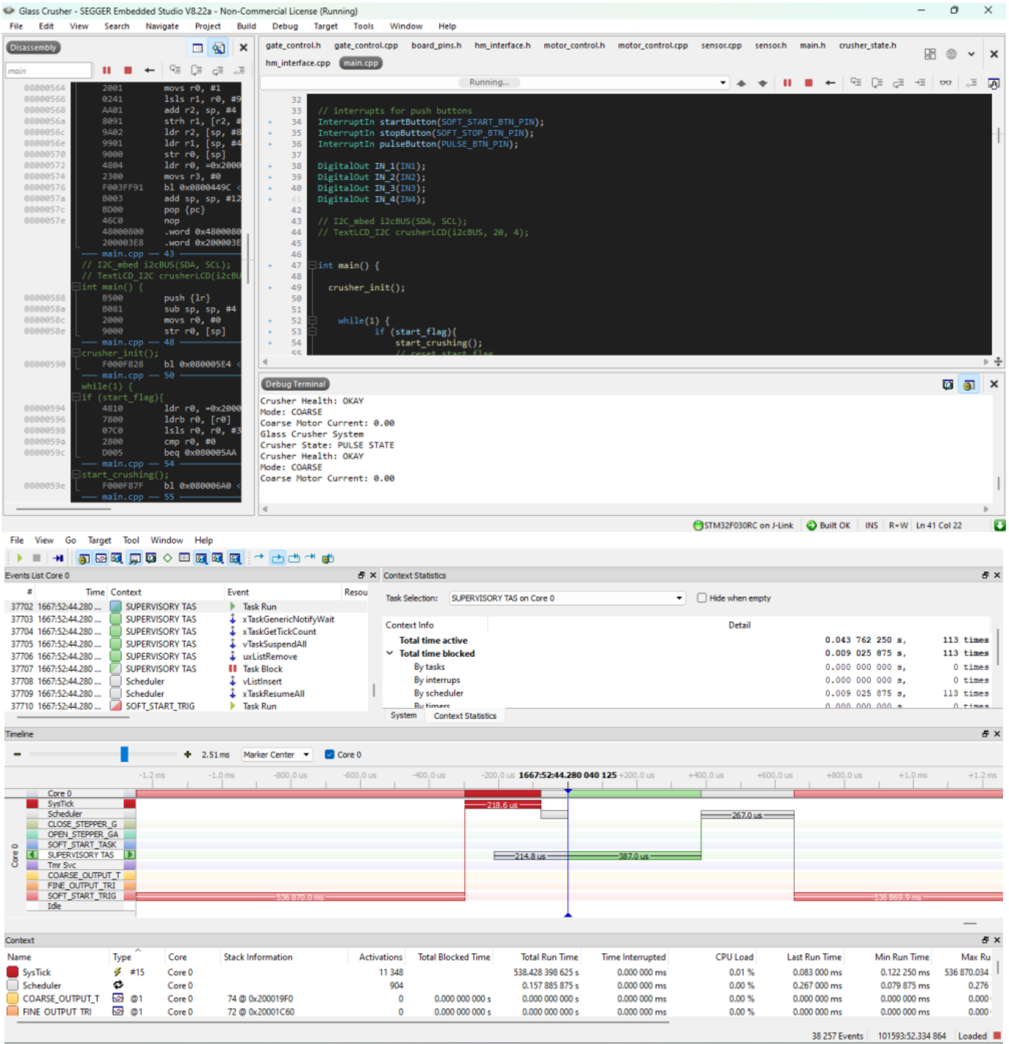

Figure 3.10: Project tree in SES (with System View for real-time analysis)

3.3.2 Algorithm Development

1. Boot Process: When the Glass Crusher System is powered on, the master microcontroller is also powered on. The system begins booting by initializing system variables, peripheral pins, and attaching all event triggered tasks to the rising edge of the Interrupt Service Routine (ISR). The system proceeds with the initialisation by attaching all periodic tasks to their respective DarkLight Ticker objects. A function named "init_tracker" tracks the whole initialisation process and displays the progress on the PC2004F LCD in real-time via the I^2C bus. A red indicator LED indicates the idle state of the system.

2. Gate Mode Operation: The Gate of the system with its intended purpose as described by the Mechanical Team in their report, cannot operate efficiently without precise control and accurate timing. Fastened to the HMI panel is a toggle switch, which allows the operator of the crusher to select the type of output. We call this Operation Mode in our design. The toggle switch measures the digital voltage level of the connected pin. The presence of a voltage (High state of digital pin) signifies a coarse crushing and the

opposite is true for the fine crushing mode. If a coarse mode is detected by the system, the master microcontroller sends motor control commands to the coarse crushing motor only. Otherwise, both crushing motors are commanded for a fine grain output of glass particles.

3. Crushing: The system initiates a soft start, soft stop or pulse crushing operation, provided an external interrupt event is triggered using the start button, stop button and pulse button respectively on the HMI. To achieve a high-performance crushing operation, and PID (Proportional-Integral-Derivative) control, the crushing motors had been given a higher priority among the various concurrent tasks enabling uninterrupted system operation during crushing. During a soft-start crushing operation, a green indicator LED (turned-on) shows the state of the system as running. For a pulse crushing operation indication, the same green LED toggles every half of a second. The LCD on the HMI panel updates the state of the system in every 500ms. Data the LCD displays include;
   a) Operation Mode.
   b) Current data of crushing motor depending on the mode of operation.
   c) The current state of the system (IDLE, STARTING, RUNNING, PULSING and PULSING).

4. Conveyor Motion: The conveyor belt's motion is manually handled by the operator using a knob attached to the HMI panel as document by the Electronics and Communication team. This gives the operator of the crusher the ability to regulate the through-put rate of the raw glass bottles.

5. PID (Proportional-Integral-Derivative): As the crushing motor attains required crushing force, the system switches to PID control mode. In this mode, the current consumption data of the rotating shaft is measured by the current sensor on the slave motor driver board. In the case of an overload, the crushing force reduces and the duty cycle of the PWM signal is adjusted to fit the required crushing force using the current consumption data.

6. Soft Stop: Soft stop is one of two (2) ways provided to bring the crushing process to a halt, through a stop button. A soft stop is initiated by an interrupt and all motors gradually come to a stop.

7. Emergency Stop: An emergency button is provided to quickly cut-off power via a relay module. This is a critical safety measure to prevent injury or damage. The emergency stop is the second way by which the crushing process comes to a halt.

8. Manual Override/Post-Emergency recovery: If the system halts mid-cycle due to an emergency, with glass still in the crushing chamber, but no bottle on the conveyor, the pulse button allows the operator to manually crush the remnants. This ensures no glass is left stuck in the crusher after a fault. The main purpose of the pulse button is to override the automatic crushing process, ensuring safe, complete crushing even after unexpected stops.

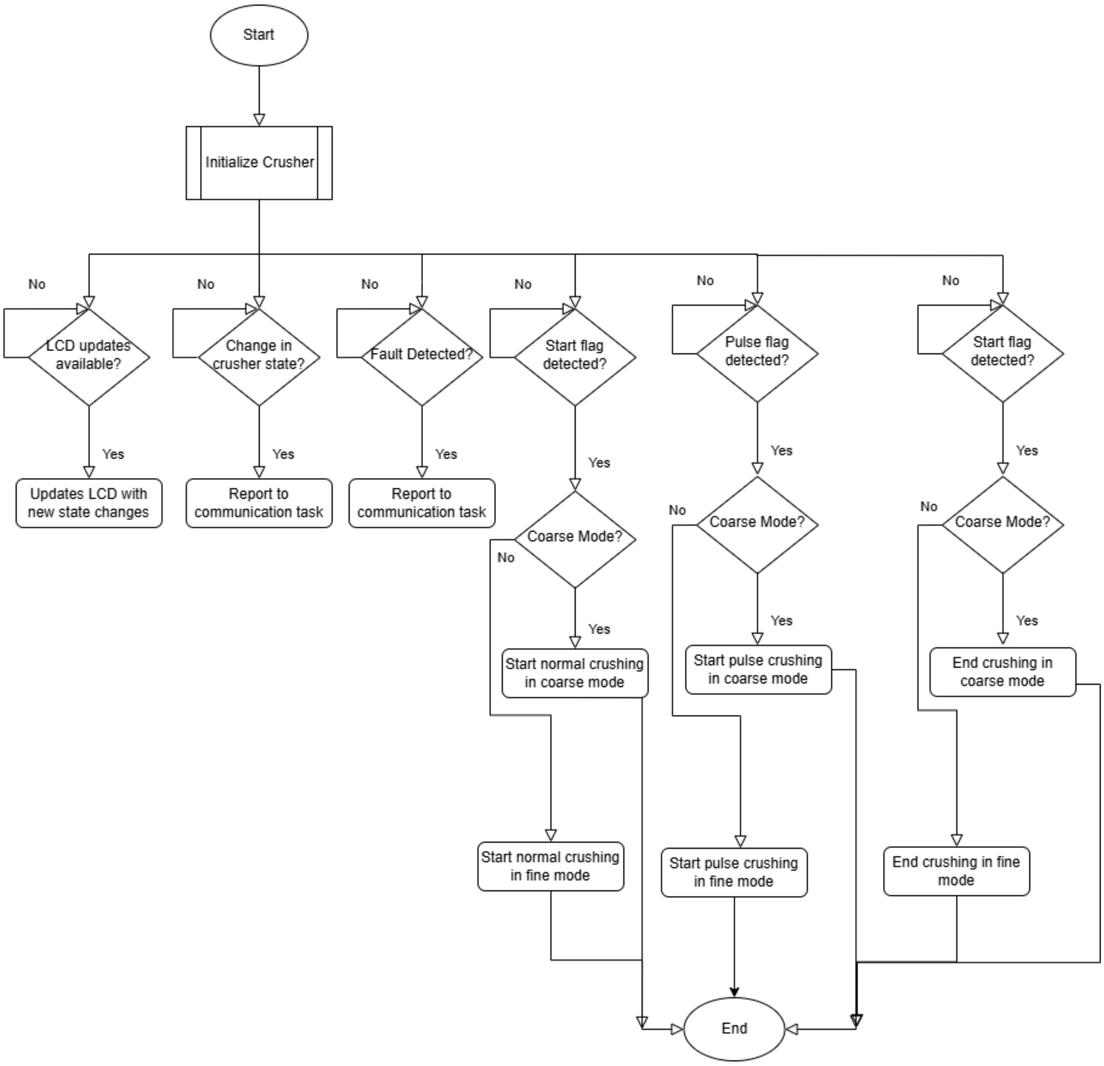

Figure 3.11: Flowchart depicting the algorithm of the crushing process

Figure 3.12 shows the control algorithm employed in the system. It highlights the integration of the Proportional-Integral-Derivative (PID) control loop and feedback from the current sensor, which ensures overload protection and quick recovery of the crushing motor.

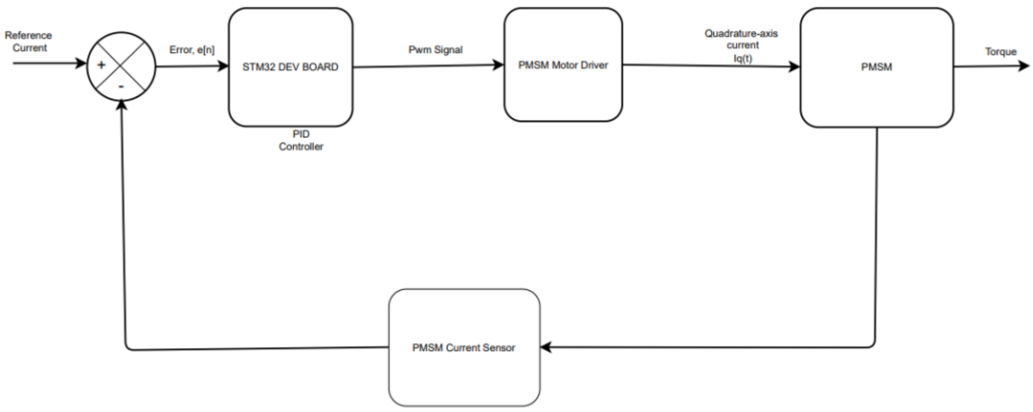

Figure 3.12: PID control algorithm

3.4 Conclusion

The methodology employed in this phase of the project combines proof-of-concept miniature prototypes for overall firmware test to minimize hardware damage on the real system, individual hardware and software tests and a later systems integration of all the individual components which will be achieved by deploying the resulting solutions and fixes onto the real system. Potential improvements such as concurrent task scheduling and master-to-slave communications on the I^2C bus will be discussed in the final section.

CHAPTER 4 : RESULTS ANALYSIS AND DISCUSSIONS

4.1 Introduction

This chapter presents the results of the entire firmware with motor control algorithm and I2C protocol implementation as an exception. The evaluation is based on the response of the system to external interrupts, gate control and gate fault detection mechanisms and how the various tasks coordinate the entire crushing process. Results are compared to theoretical expectations to determine the performance of the hardware set-up and the underlying firmware.

4.2 Artefact Result Discussion

4.2.1 Unit Test:

Unit tests of some vital functions were performed using the unity testing framework to verify code quality and detect deficiencies. The following functions were tested individually and verified:

HMI Functions:

hmi_init (), hmi_update_lcd (), hmi_set_green_led (), hmi_flash_green_led ()

hmi_set_red_led (), hmi_flash_red_led (), hmi_beep (), hmi_stop_buzzer ()

get_crusher_mode (), start_button_handler (), stop_button_handler ()

pulse_button_handler ()

Motor Control Functions:

start_motor (), stop_motor (), pulse_motor (), check_motor_fault ()

get_motor_current_from_coarse_slave (), get_motor_current_from_fine_slave (),

Gate Control Functions:

open_gate (), close_gate (), is_gate_opened (), is_gate_closed (), get_gate_state()

State Machines:

handle_crusher_state ()

4.2.2 Human Machine Interface (HMI) Panel:

The master microcontroller was powered via the ST-Link V2 debugger. The following initialisations were successfully observed and recorded:

a) The red LED was initially turned-on while the LCD displays the progress of the initialisation process as indicated in Figure 4.1 and 4.2. This process was not timed because the initialization time of the system varies on every boot process. Averagely the boot process takes about 3 to 5 seconds to complete. Non-latching momentary push buttons were used to perform start/stop/pulse operations. The interrupt handlers run as expected and the various flags are updated according to the respective button pressed.

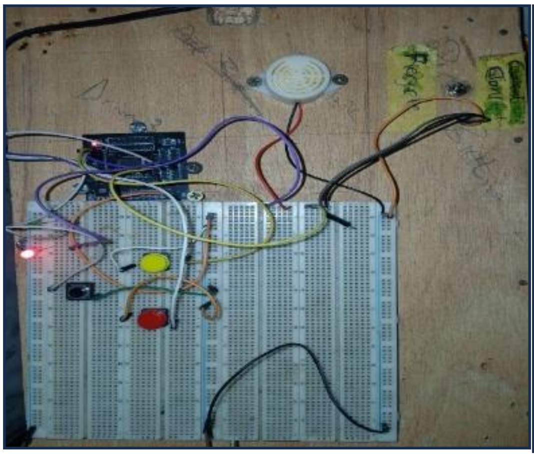

Figure 4.1: 20x4 LCD showing boot process

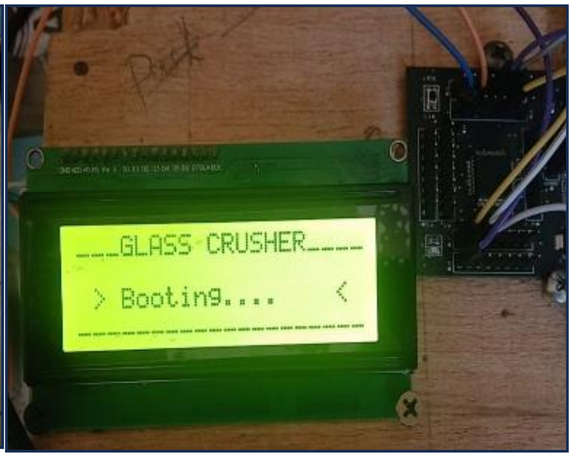

Figure 4.2: Red LED turned on showing idle state of crusher

4.2.3 Gate Control:

The gate (diverter) was rigorously tested and observed for precise opening and closing operations.

A Nema17 stepper motor was attached to a 3D printed bar not weighing more 20 grammes was used for testing purposes. Two limit switches were attached to the ends of the test board where the resting position of the gate is either opened or closed as shown in Figure 4.3. Initially the gate is positioned on the closed state limit switch. In this state, the inner-outlet of the coarse crushing chamber leading into the fine crushing chamber is blocked or closed by the gate allowing any crushed glass particles to fall freely out of the coarse crusher. When the mode changed to fine, several observations were made. The gate precisely moved from the closed limit switch and rested on the opened limit switch within a time frame of 2-3 seconds confirming a successful opening and closing of the gate. The mode of the gate can only be updated after a stop has been initiated and a new start or pulse operation is about to begin. A final test on the gate was done by stopping the bar from reaching either of the limit switches. This caused the red LED to blink and the buzzer emits a continuous beep sound until the error was fixed by allowing the gate to fully close/open. This is just one of two error detection mechanisms implemented for the gate control.

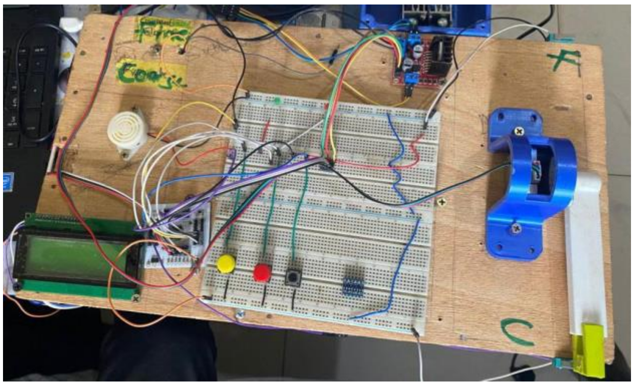

Figure 4.3: Test board with Nema17 Stepper motor and attached 3D printed bar

4.3 Implications for Future Work

4.3.1 Potential Enhancements

Based on the analysis, several enhancements could be made to improve the system. These include:

1. Fault Detection Mechanisms: Several fault issues may arise as a result of continues usage of the crusher. To safely detect and resolve these potential faults, few software fault detection mechanisms were implemented for the Gate and the Motors but only few were tested. For a robust Glass Crusher system, component specific faults should be detected and reported to the operator via the LCD display. This helps improve the reliability of the system by optimizing maintenance.
2. Motor Control Algorithm: A PID control algorithm is recommended for the efficient operation of the system as discussed in the previous chapters. This should be implemented in future enhancements of the project.
3. Others: Other safety protocols like the emergency stop and software interlocks which were discussed in the previous chapters but not implemented as an objective in this project should also be considered in further development works.

4.4 Conclusion

The results analysis demonstrates that the Glass Crusher system achieved its design objectives with regards to the software implementation side of things as outlined in chapter 1 of this report.

The results of this project shows that the three (3) outlined objectives, which include designing of a user interface, implantation of crusher operations (start/stop/pulse) and a proof-of-concept test were achieved. While there are areas for potential improvement, the overall performance validates the system design and the methodology used. The insights gained from this project can inform future works in firmware development and related fields, paving the way for further innovation and development.

CHAPTER 5: SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The Motor Control and Software Implementation project successfully achieved its objectives of developing and implementing a firmware for the Glass Crusher System. By integrating event-triggered and time-triggered software architecture based on individual task implementations, we designed a robust and easy to maintain and improve Glass Crusher Firmware. The project demonstrated how the entire operation of the physical crusher can be grouped into individual tasks like motor control tasks, gate control tasks and HMI tasks, and be assigned priorities based on external events and time deadline events. With components such as the toggle switch enabling the switching of the two different modes of operation, the limit switches for detecting the successful closing/opening of the gate and as well as errors, indication lamps and alerts and as well as displays to convey system information to the user. Moreover, the hardware and software were integrated on a test board which proves the overall behaviour and performance of the system and wouldn't differ in any case if integrated onto the Glass Crusher itself. Throughout the testing stage, the system responded effectively to various input commands and forced error tactics, demonstrating the robust design and effectiveness of the implemented software components. The results validated the initial choices of hardware and software components. This project contributes significantly to the field of embedded systems and as well as sustainable development. It provides a scalable, robust, and cost-effective firmware solution that can be adapted for broader applications in Glass Crusher Systems.

5.2 Recommendations

Based on the project's findings, several recommendations are suggested to enhance future implementations and address the limitations observed:

1. Sensors: Temperature and vibration sensors as recommended components could be used to monitor the system's performance and detect potential faults due to overheating and abnormal vibrations that might occur in the course of operating the crusher for a longer duration of time.
2. Real-Time Operating System (RTOS): The firmware solution for our Glass Crusher system uses the traditional infinite loop approach and external interrupt handlers to execute higher priority tasks like the motor control task. The loop approach is where each component of the application is represented is represented by a function that executes to completion. Ideally a hardware timer would be used to schedule the time critical motor control function. However, having to wait for the arrival of data and the complex calculation performed make the motor control function unsuitable for execution within an Interrupt Service Routine (ISR). Even though the motor control functions are executed outside of the ISR using flags updated inside the ISR, the updating of the flags also take a bit of time as the processor pauses execution of the control algorithm while doing so. The simple loop approach is very good for small applications with flexible timing requirements but can become complex, difficult to analyse and maintain if scaled to larger systems like the glass crusher. With these limitations observed, we are recommending a fully pre-emptive multitasking solution which makes use of RTOS services with no regard to the resultant memory and processor overhead. This approach is simple, segmented, flexible and maintainable.

The event driven structure of such systems ensures that no CPU time is wasted polling for events that have not occurred. Processing is only performed when there is work needing to be done.

This project successfully lays the groundwork for advancing both sustainable development and Glass Crusher software applications. The implemented software has already proven to be a scalable and effective tool. By acting on the provided recommendations, subsequent developments can yield Glass Crusher Machines with significantly greater sophistication and operational efficiency.

References

[1] O. T. Tesfaye and A. T. Getaw, "Design of impact stone crusher machine," *International Journal of Scientific and Engineering Research,* vol. 10, no. 5, p. 6, 5 May 2019.

[2] W. Bank, "Environmental, Health, and Safety Guidelines: GLASS MANUFACTURING," World Bank, 30 April 2007. [Online]. Available: https://documents1.worldbank.org/curated/en/890101490072833164/pdf/113621-WP-ENGLISH-Glass-Manufacturing-PUBLIC.pdf. [Accessed 1 May 2025].

[3] G. Christina , W. Ernst , M. Eric and H. C.-G. Wina , "Energy Efficiency Improvement and Cost Saving Opportunities for the Glass Industry. An ENERGY STAR Guide for Energy and Plant Managers," *An ENERGY STAR® Guide for Energy and Plant Managers,* p. 113, 6 March 2008.

[4] R. K. Muhammad, A. K. Aleem and G. Umer, "Speed Control of DC Motor under Varying Load Using PID Controller," *International Journal of Engineering (IJE),* vol. 9, no. 3, p. 11, 2015.

[5] T. Rob and . W. Tim, "Fast and Effective Embedded Systems Design: Applying the ARM mbed," in *Fast and Effective Embedded Systems Design: Applying the ARM mbed*, Cambridge, Newnes Publications, 2017, p. 512.

[6] A. Kwame , A. C. Mark , M. D. Ocansey , D. Enock , K. O. Thomas and O. Gideon , "Design and Fabrication of a Vehicle Access Control System for the Entrance of Koforidua Technical University : Motor Control and Software Implementation," None, Koforidua, Ghana, 2024.

[7] P. Harshvardhan, P. Rohit, P. Manish, J. Tejas and C. Avinash, "Design and Fabrication of a Glass Crushing Machine," EasyChair Preprint, Palghar, India, 2020.

[8] I. A. Md., I. Naeemul , H. R. Md., . M. T. R. S and S. H. Md., "Design a Real Time Temperature Monitoring System to Protect DC Motor," in *22nd International Conference on Computer and Information Technology (ICCIT)*, Chittagong, Bangladesh, 2019.

[9] B. Baker, "EDN," Baker's Best Blog, 22 September 2011. [Online]. Available: https://www.edn.com/designing-with-temperature-sensors-part-one-sensor-types/. [Accessed 5 April 2025].

[10] V. O. R. João , P. G. P. Marcos , F. . B. S. Flávio and G. L. J. Reginaldo , "Performance evaluation of accelerometers ADXL345 and MPU6050 exposed to random vibrational input," *Research Society and Development,* vol. 10, no. 15, p. 16, 2021.

[11] B. EMEA, "Sensors for Detecting Transparent Media : Innovative Solutions," Balluff EMEA, 05 12 2024. [Online]. Available: https://www.balluff.com/en-se/blog/sensors-for-detecting-transparent-media-innovative-solutions-from-balluff. [Accessed 20 05 2025].

[12] T. Rob and W. Tim, Fast And Effective Embedded System Designs: Applying the ARM mbed, vol. 2nd Edition, Cambridge: Jonathan Simpson, 2017, p. 512.